

Set 19: Intermolecular forces I

Notes

Observable properties of covalent molecular substances can be explained by the type and strength of intermolecular forces present between the molecules.

Dispersion forces are present between all molecules and are caused by temporary dipoles created by transitory imbalances in the electron cloud around a molecule. These temporary dipoles then induce dipoles in neighbouring molecules. The strength of dispersion forces are primarily determined by the number of electrons in the molecule, that is, the larger the number of electrons, the greater the strength of the dispersion forces. The surface area of the molecules will also impact on the strength of the dispersion forces. The greater the surface area, the greater the dispersion forces. This explains why isomers, with the same number electrons, have different boiling points.

Dipole-dipole forces of attraction will occur between polar molecules in addition to the dispersion forces. When comparing polar and non-polar molecules with similar numbers of electrons, differences in properties such as boiling point, can be explained by the presence of the additional dipole-dipole forces.

Hydrogen bonding is an extreme form of dipole-dipole forces. Where a very polar covalent bond between hydrogen and a very electronegative element such as fluorine, oxygen or nitrogen, exists in a molecule, the electron density around the hydrogen will be very low. As a result, the hydrogen atom will have a relatively high positive charge. The low electron density also allows the hydrogen atom to approach an unbonded pair of electrons on a neighbouring molecule's oxygen, nitrogen or fluorine (which are all relatively negatively charged) more closely. This combination of close proximity and relatively large difference in charge causes the dipole-dipole force of attraction to be much stronger than usual. This explains the higher than expected melting and boiling points in substances such as water, ammonia and hydrogen fluoride.

Set 19: Exercises

1. For the following substances, identify the type(s) of intermolecular forces present between the molecules.
 - (a) octane (C_8H_{18})
 - (b) water (H_2O)
 - (c) ethanol ($\text{C}_2\text{H}_5\text{OH}$)
 - (d) pure nitric acid (HNO_3)
 - (e) glucose ($\text{C}_6\text{H}_{12}\text{O}_6$)
 - (f) iodine (I_2)
 - (g) hydrogen chloride (HCl)
 - (h) methane (CH_4)
2. List the following substances in order of increasing boiling point and justify your choices.
 - (a) HF , HI , HCl , HBr
 - (b) H_2O , NH_3 , O_2 , CH_3Cl

3. Given the following substances and their boiling points, list the substances in order of increasing vapour pressure at room temperature. Justify your choices.

Substance	Boiling point ($^{\circ}\text{C}$)
H_2O	100
HNO_3	83
Octane	125
Mercury	357

4. Predict the solubility of each of the following substances in water and explain your decisions.
- (a) Methane (CH_4)
 - (b) ethanol ($\text{C}_2\text{H}_5\text{OH}$)
 - (c) ammonia (NH_3)
 - (d) hexane (C_6H_{14})
 - (e) iodine (I_2)
 - (f) glucose ($\text{C}_6\text{H}_{12}\text{O}_6$)
5. Predict the solubility of each of the following substances in hexane (a non-polar liquid) and explain your decisions.
- (a) Methane (CH_4)
 - (b) ethanol ($\text{C}_2\text{H}_5\text{OH}$)
 - (c) ammonia (NH_3)
 - (d) hexane (C_6H_{14})
 - (e) iodine (I_2)
 - (f) glucose ($\text{C}_6\text{H}_{12}\text{O}_6$)

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